

Workflow reliability — Design

AC01–AC09 · Base bcc971d · Revision 2 · Proposed, not implemented

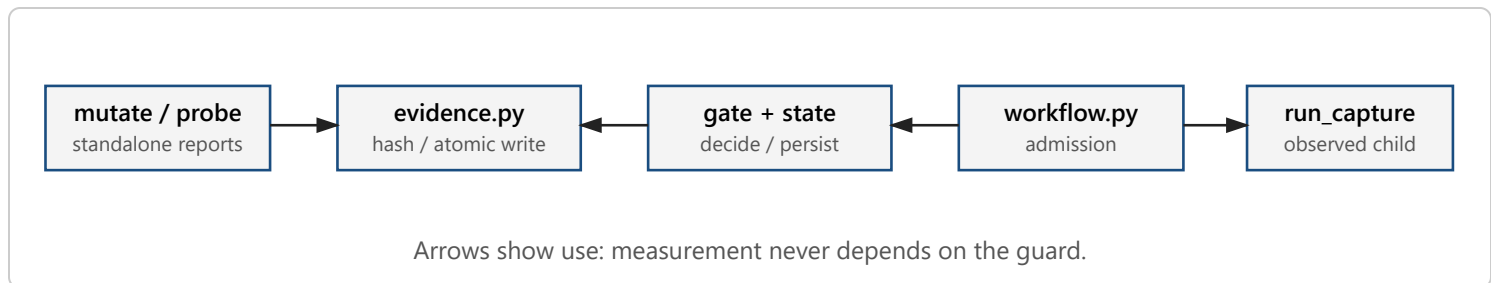
Inputs: [scope](#), [accepted research](#), [traceability](#) · [Complete interfaces, records and invariants](#)

1. Problem & scope

FACT: [Parent-run, source-pinned counterexamples](#) confirm omitted ESM/CJS changes, a syntax-invalid mutant scored as a 100% kill, unavailable mutation reported pass, and stale-score reuse after failure. These are pre-fix failures, not feature passes. Workflow obligations remain prose (research E16–E18). Preserve dependency-free Python/Node CLI/docs; no application UI, framework or host API.

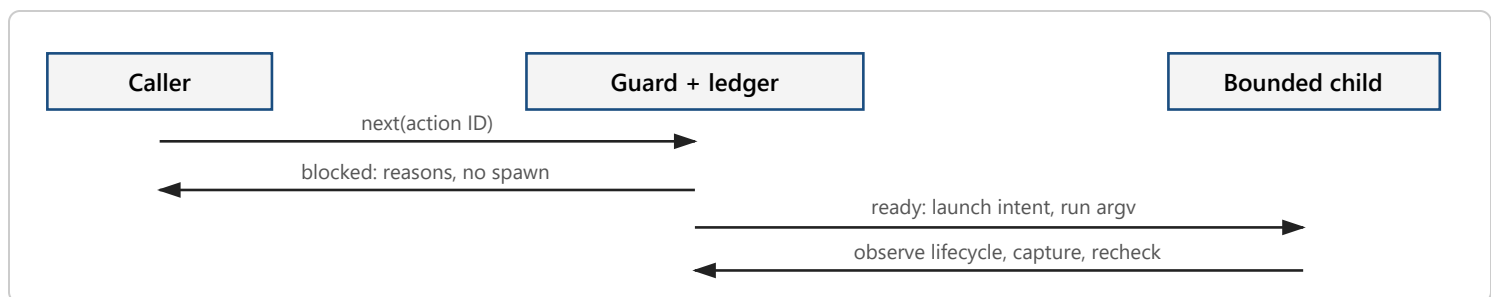
Human approval remains unconfirmed under authorized autopilot. Accepted DR1–DR7 dispositions are in the linked contract. I1 measurement works without adoption or ledger; I2 adds guarded dispatch/resume; I3 integrates references and final proof. Separate planning owns tasks and scheduling.

2. Program design



Component / AC	As-is → proposed interface	Reuse / responsibility
Measurement 01, 02, 09	<code>mutate.py</code> : <code>select_test_run(repo, cwd, argv)</code> ; <code>probe.py</code> : <code>assess_report(metrics, policy)</code> . Exit-only scoring → fresh, classified proof.	Extend discovery, operators, <code>judge</code> , <code>mutation_entry</code> (E01–E10). Keep collection separate from policy.
Native results 01, 06, 09	New <code>native_result.mjs</code> : <code>report(source)</code> , <code>parseReport(text)</code> . Shared Node reporter for mutation and <code>runTestQuality</code> .	Replace assertion-string/count heuristics (E12/E19); native test events, not a new runner.
Readiness 03, 05, 07, 08	New <code>workflow_gate.py</code> : <code>evaluate(contract, ledger, inputs, target)</code> → Readiness.	Pure decisions (E16–E18). Action/closure <code>requires</code> own edges; reverse/due/replay views are derived.
Durability 01–05, 07	New <code>evidence.py</code> : <code>source_snapshot</code> , <code>write_json_atomic</code> , <code>exclusive_lock</code> . Later <code>workflow_state.py</code> : <code>bind_inputs</code> , <code>append_event</code> .	Standalone file primitives first; guard-specific contract binding later. No ledger needed to emit metrics (E19).
Dispatch / status 01, 04, 05, 08, 09	New <code>workflow.py</code> <code>main(argv=None)</code> ; six verbs. Extend <code>run_capture</code> with optional <code>env</code> and lifecycle <code>observe(event)</code> .	Expose actual child ownership while alive (E05); persist launch intent and observations. Unknown recovery stays blocked.
Workflow / evaluation 04–09	Current-design history, earlier compatibility, owned corrections, derived status; native protocol tests and separate parent-host exercises.	Wire SKILL plus delegation/testing briefs (S13), not a new host or evaluation framework.

3. Key interactions



`python scripts\workflow.py next --slug workflow-reliability --action CHECK-ID` selects only registered `argv/cwd`. Status returns this controlled command only when ready. A working command is not completion: closing an increment additionally requires due independent verification/review and prerequisite completion.

4. Data & contracts

`SourceSnapshot` needs no workflow. Later `ContractBinding` explicitly adds current component/check hashes. `workflow.json` owns forward edges/commands; `current-design.json` points to history; the ledger owns runtime facts. Replay/status are derived. Own generated outputs do not invalidate source proof.

Metric v2 separates measured regression from completeness. Unknown baseline is not greenfield; stale reports cannot become new measurements. Missing proof fails without imports, exemptions or adoption. Mutation stays 60%; eval stays 0.9. Known-greenfield cycles fail under the existing no-new-cycle rule.

5. Decisions & failure behavior

- **Explicit argv, not shell text:** preserve spaces and package cwd independently of Git root. Node events must identify relevant assertions; syntax, setup, empty suites and timeouts never kill mutants. Unsupported runners remain unavailable.
- **Observed ownership, not a supervisor:** persist launch intent and actual child events. The spawn-to-record crash gap remains unknown; dead guard/PID age cannot unlock it. Recovery needs supported termination evidence, otherwise stays blocked.
- **Observed scope, not sandbox:** hashes detect only scoped input changes. Review owns scope completeness. Identity is self-declared; outside-scope writes remain unobserved. Keep owned scratch cleanup, without a product-wide OS-temp ban.
- **Scoped reopening, not full reset:** changed prerequisites invalidate dependent proof only. Old red/compatibility proof remains historical; retirement requires its admission before deletion, never a later backfill.
- **Fail closed:** blocked gate exits 1, malformed input 2, lock/recovery conflict 3. Executed children preserve exit/timeout codes and carry a separate structured outcome. Approval is not inferred from silence.

6. Test strategy — proposed, not run

ACo1–o2: real ESM/CJS repositories without workflow artifacts; spaced argv/nested cwd; classifications and stale-report failure. **ACo3–o5:** no-spawn sentinel, forward-edge rules, live-child guard crash, uncertain spawn, recovery evidence and scoped resume. **ACo6–o8:** failure evaluations, temporal retirement, current delegated contract and research supersession. Separate fresh-agent diagnosis withholds cause/oracle from neutral inputs and records available accesses; shared storage is not isolation. Exposure or insufficient records leaves blindness unverified. **ACo9:** independent per-increment verification/review and final native/eval regressions. Required missing proof remains incomplete, never waived.